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Optimal Operation and Services Scheduling for an Electric Vehicle Battery Swapping Station

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***Abstract*—For a successful rollout of electric vehicles (EVs), it is required to establish an adequate charging infrastructure. The adequate access to such infrastructure would help to mitigate concerns associated with limited EV range and long charging times. Battery swapping stations are poised as effective means of eliminating the long waiting times associated with charging the EV batteries. These stations are mediators between the power system and their customers. In order to successfully deploy this type of stations, a business and operating model is required, that will allow it to generate profits while offering a fast and reliable alternative to charging batteries. This paper proposes an optimization framework for the operating model of battery swapping stations. The proposed model considers the day-ahead scheduling process. Battery demand uncertainty is modeled using inventory robust optimization, while multi-band robust optimization is employed to model electricity price uncertainty. The results show the viability of the proposed model as a business case, as well as the effectiveness of the model to provide the required service.**

***Index Terms*—Battery swapping station, electric vehicles, energy storage, robust optimization.**

NOTATION

The notation used throughout the paper is stated below for quick reference.

Sets:

B	Set of multi-band robust optimization uncertainty bands with index b.
D	Set of discount segments with index d.
G	Set of battery capacity groups with index g.
I	Set of batteries with index i.
J_g	Set of time periods in which battery demand uncertainty may materialize for battery group g.
T	Set of time periods with index t.
U_b	Set of time periods in which electricity price uncertainty may materialize for band b.

Continuous Variables:

$bat_{i,t}^{\mathrm{chg}}$	Charging power of battery i at period t (kW).
$bat_{i,t}^{\mathrm{dsg}}$	Discharging power of battery i at period t (kW).
$C_{i,t}^{\mathrm{deg}}$	Degradation cost for battery i at period t (\$).
em_t^{buy}	Energy purchased in the wholesale market at period t (kWh).
em_t^{sell}	Energy sold in the wholesale market at period t (kWh).
$L_{i,t,d}$	Energy shortage with respect to the discount curve for battery i in each discount segment d at period t (kWh).
q_g	Auxiliary variable for linearity in battery group g.
$r_{g,t}$	Auxiliary variable for calculation of demand uncertainty in group g at period t.
$soc_{i,t}$	Energy state-of-charge for battery i in period t (kWh).
$soc_{i,t}^{\mathrm{short}}$	Shortage of energy in battery i when being swapped at period t (kWh).
v_b	Auxiliary variable for linearity in band b.
y_g	Penalty cost for battery demand uncertainty (\$).
z_t	Auxiliary variable for calculation of electricity price uncertainty at period t.
$\beta_{i,t}$	Discount percentage given to the customer swapping for battery i with energy shortage at period t.

Integer Variables:

$bat_{g,t}^{\mathrm{short}}$	Number of unsupplied batteries for group g in period t.
$bat_{g,t}^{\mathrm{add}}$	Number of additional batteries for group g in period t.

Binary Variables:

$a_{i,t}$	Auxiliary variable for battery i at period t.
c_t	Auxiliary variable at period t.
$x_{i,t}$	Swapping status of battery i at period t (1 if swapped, 0 otherwise).

Parameters:

BC_g	Nominal battery capacity for group g (kWh).

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$BC_g^{\max}$	Maximum battery capacity for group g (kWh).
$BC_g^{\min}$	Minimum battery capacity for group g (kWh).
BSR	Revenue obtained from customers for a battery swap ($).
C_i^{bat}	Cost of battery i ($/kWh).
k_d	Slope of discount curve segment d (%/kWh).
$L_d^{\max}$	Maximum limit of each state-of-charge shortage segment d (kWh).
m_i	Linear slope approximation of the life of battery i as a function of number of cycles.
M	Large integer value.
N^{MC}	Number of trials.
$N_{g,t}$	Expected battery demand for group g at period t.
$N_{g,t}^{\max}$	Maximum battery demand for group g at period t.
$N_{g,t}^{\min}$	Minimum battery demand for group g at period t.
$P_i^{\max}$	Maximum power of battery i (kW).
$S_{i,g}$	0/1 parameter indicating if battery i belongs to group g.
$SOC_{i,t}^{\text{init}}$	Initial (for $t=0$) and incoming (for $t>0$) energy state-of-charge of battery i at period t (kWh).
$VoCD$	Value-of-customer-dissatisfaction ($).
Γ_g	Robustness parameter for battery demand in group g.
$\Delta N_{g,t}$	Battery demand deviation for group g at period t.
$\Delta\lambda_{t,b}$	Market price deviation in band b for period t ($/MWh).
η^{chg}	Battery charging efficiency percentage.
η^{dsg}	Battery discharging efficiency percentage.
θ_b	Robustness parameter for market prices in band b.
λ_t^{DA}	Expected market price at period t ($/MWh).
$\lambda_{t,b}^{\max}$	Maximum market price in band b at period t ($/MWh).
$\lambda_t^{\min}$	Minimum market price at period t ($/MWh).

I. Introduction

ELECTRIC vehicles (EVs) are being widely adopted not only because they reduce reliance on fossil fuels, but also because their operation is less carbon intensive than their counterparts that use internal combustion engines. However, potential customers are still deterred to purchase EVs due to the associated long battery charging times, limited range and high battery replacement cost. An efficient solution to these problems involves the deployment of battery swapping stations (BSS). In these stations, drivers can swap their discharged battery with a charged one. From the power system perspective, the BSS is then a large flexible demand, and from the EV owner's perspective, it is a service provider that can supply them with a fully charged battery on request at a fee. This service is similar to the service that gasoline stations provide to internal combustion vehicles.

The large stream of benefits that the EVs offer to power systems, such as voltage support [1], frequency regulation, reserve [2] and demand response capabilities [3], have been the subject of several research works. A constant over all these approaches is that an aggregation entity that operates the EVs as ensembles is required in order to have measurable impacts on the power system. Since the BSS is a new player that aggregates and operates a large number of EV batteries, the similar benefits can be obtained from the BSS without the need of a mediator. As an objective, the BSS seeks to maximize its profits, by participating in markets and providing services, such as demand response, energy storage, and reserves. The storing capabilities of the BSS are scheduled based on time-varying electricity prices, i.e., the real-time pricing (RTP) [4]. The BSS maximizes its profits by exploiting the low-price periods of the day to purchase electricity and charge batteries in Grid-to-Battery mode (G2B), and sell during the high-price periods by discharging batteries to the grid in Battery-to-Grid mode (B2G). Additionally, BSS can perform Battery-to-Battery (B2B) services in order to charge certain batteries using the energy stored in other batteries. Several works in the literature are focused to similar services provided directly by EVs. These are: Grid-to-Vehicle (G2V), Vehicle-to-Grid (V2G), and Vehicle-to-Vehicle (V2V) [5]. Therefore, G2B, B2G, and B2B modes introduced in this paper are BSS counterparts of EV's G2V, V2G, and V2V modes, respectively.

Although the BSS concept is novel, there are already some pioneer works in the field of its planning and operation. In [6], the optimal locations where BSS can be installed and operated in distribution systems are determined. In this model, the type of load, the required reinforcements to the distribution system, and reliability of the system are explicitly considered. However, the EV model uses a heuristic approach to determine charging/discharging schedules. An economic dispatch model that uses BSS to manage wind power intermittency is developed in [7]. In [8], the number of batteries to be purchased along with their charging schedules are determined using a basic dynamic programming framework. However, the number of batteries purchased depends on the scheduling model of the EV batteries which, in such an approach, is simplified to a wide extent. In all the works above, the interaction with the electricity markets is ignored. The business aspect of the BSS has also been the subject of research. The idea of a subscription pricing structure, along with the required infrastructure cost, is presented in [9]. The associated risks, classification of investments, and potential services that could be sold by the BSS are investigated in [10]. The detailed cost analysis required for the startup of a BSS is performed in [11]. However, these models are simplified since they do not consider the interactions between the BSS and power system.

The BSS business and operational models have not only been treated in theory. Commercial businesses have developed around the BSS concept to take advantage of the existing EV populations. For example, the company Better Place in 2012 installed multiple stations that handle specific type of EVs [12]. In 2013, Tesla Motors introduced battery swapping technology for their EVs [13]. Also, several utilities in China installed BSSs for their EV population in 2013 [14]. However, the profits in such actual BSSs are entirely dependent on the fees charged for battery swapping, and ignore the extra revenue that could be collected by participating in the markets.

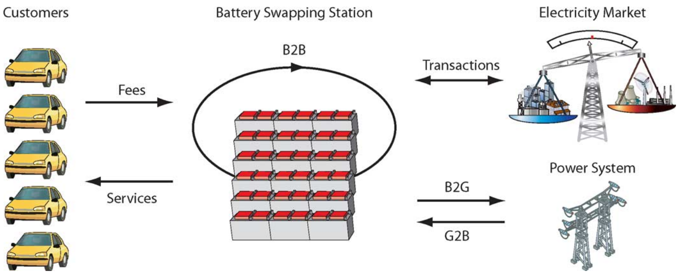

Fig. 1. BSS interactions with customers, market, and the power system.

The work in this paper is a major overhaul of the approach presented in [15]. The optimization model incorporates the robust day-ahead (DA) decision stage where the BSS can hedge against market price and battery demand uncertainty. In this stage, the optimal battery schedules are determined. In particular, the multi-band robust optimization (RO) [16] is applied to hedge against price uncertainty, and inventory-theory RO [17] is applied to accommodate the uncertain demand of batteries. The models consider basic EV characteristics, expected customer behavior, and cost of battery degradation.

The main contributions of the paper are:

- A realistic framework of the DA scheduling process in order to take advantage of G2B and B2G services.
- Complete BSS operating model including battery degradation, market price uncertainty, and battery demand uncertainty. The impact of each feature is individually analyzed.
- The model exploits the ability to transfer energy among batteries in B2B mode, if there is an economic benefit.

The rest of the paper is organized as follows. Section II describes the BSS business model. Section III proposes an optimization model for the BSS and Section IV discusses the results. Finally, Section V presents conclusions.

II. Business Case

A. Operations

The operation of the BSS is shown in Fig. 1 [15]. The BSS's goal is to supply the battery demand while maximizing its profits. The BSS requires a stock of batteries with different capacities to serve its customers. It is assumed the batteries are owned by the BSS and leased to the customers. The customers benefit from this arrangement since all the costs related to the batteries, including degradation and maintenance, are accrued by the BSS. The customer is not concerned with the battery lifetime nor with the way in which the battery was charged (rapid, high power charging or slow charging). To recoup these costs and to make a profit, the BSS charges customers a fixed fee for the service of swapping.

The BSS decides the optimal battery charging/discharging schedules and purchases/sells electricity by submitting bids and offers in the electricity market. By exploiting the time-dependencies of wholesale electricity prices, the BSS can take advantage of buying electricity (in G2B mode) during the low-price periods, selling electricity (in B2G mode) during the high-price periods, and shifting energy within its stock of batteries (in B2B mode). To the authors' best knowledge, this is the first introduction of the B2B concept in the literature.

As an example of B2B mode, consider a scenario of two 24 kWh batteries with current state-of-charges 18 kWh and 21 kWh, in which one of them is to be swapped in the near future, and the price of electricity is high. If we assume the power rating is 3 kW, and ignore the efficiencies for the sake of simplicity, then the BSS has three choices: it can purchase 3 kWh in the market and charge the 21-kWh battery (G2B mode), it can purchase 6 kWh in the market and charge the 18-kWh battery (G2B mode), or it can discharge 3 kWh from the 18-kWh battery to fully charge the 21-kWh battery (B2B mode). Since the electricity price is high, it is more economical to perform B2B and thus postpone G2B to low-price time periods. The proposed optimization model takes advantage of these scenarios in order to maximize profits for the BSS.

B. Customer Perspective

From the customers' perspective, the largest cost of owning an EV is the battery. In 2012, the replacement cost of a 24 kWh battery ranged from \$12 000 to \$14 400 (i.e., 500–600 \$/kWh) [18]. However, in the proposed BSS scheme, the customers would lease the battery from the BSS and avoid a lump investment.

The other aspects that concern potential EV owners are the long charging times, the costs of upgrading household installations to high power chargers, and the limited number of public charging stations. Typical EVs are equipped with standard 1.6-kW level I chargers that connect directly to a household outlet [19]. At this charging level, it would take about 15 h to fully charge a 24-kWh battery. Owners, however, can install upgraded level II chargers [19], with a power level of 3.3 kW and thus reduce the charging time to about 7 h. However, the upgraded charger comes at an extra cost, which is approximately \$849 in 2013 [20]. Even if these additional expenses are acceptable to the owners, they may not be able to install upgraded chargers due to limits in their current electric supply installations, or the regulations of the housing structure they reside in (e.g., apartments).

Another concern of the EV owners is the limited range due to the relative small capacity of the batteries. In order to ease this concern, the owners would need to have access to public charging stations, which are translated into requiring heavy infrastructure investments. These concerns could be eliminated if an EV owner has access to BSSs in the areas where they usually travel.

C. Power System Benefits

As the penetration of EVs increases [21], and if instead of resorting to the BSS, the consumers favor high-power charging, they would need to install upgraded chargers to accommodate their daily energy needs for transportation. This may require upgrades not only of the electric installations at the household level, but also of the distribution system itself, in order to successfully accommodate the increased power demand. On the other hand, if BSSs are installed, some of the investments would be avoided, and the only required upgrades would be at the site at which the BSSs are located.

Furthermore, the BSS is an aggregator of batteries, and these stations could also be used to provide services to the system as a whole. The BSS can inject power back into the power system to smooth the net daily demand curve, if the BSS perceives a benefit in doing so. In addition to acting as a storage device, the BSS can also provide a share of the required ancillary services in different intervals, e.g., frequency regulation, load following, and voluntary reserve provisions.

III. Optimization Model

A. Assumptions

In the DA scheduling, the BSS estimates the market price λ_t^{DA} and the battery demand $N_{g,t}$. With this information, the DA optimization takes place. In this process, the discounts for swapping only partially charged batteries to the customers are incorporated. These discounts are a form of compensation for the inconvenience caused to the customers, and can be adjusted based on the BSS business. The DA model determines the battery charging, discharging and swapping schedules as well as the offering/bidding schedule to the market.

B. Day-Ahead Model

In the DA, the model determines the amount of electricity to buy em_t^{buy}, and to sell em_t^{sell}, in the market to meet the battery demand $N_{g,t}$. An optimal charging/discharging schedule is derived to maximize profits for the BSS. The DA model is formulated as

$$\text{maximize} \quad BSR \sum_{(t\in T)} \sum_{(i\in I)} x_{i,t} - \sum_{(t\in T)} \lambda_t^{\mathrm{DA}} \left(em_t^{\mathrm{buy}} - em_t^{\mathrm{sell}} \right) - VoCD \sum_{(t\in T)} \sum_{(g\in G)} bat_{g,t}^{\mathrm{short}} - BSR \sum_{(t\in T)} \sum_{(i\in I)} \beta_{i,t} \tag{1}$$

subject to

$$soc_{i,t} = \left(soc_{i,t-1} + bat_{i,t}^{\mathrm{chg}} \eta^{\mathrm{chg}} - \frac{bat_{i,t}^{\mathrm{dsg}}}{\eta^{\mathrm{dsg}}} \right) \cdot (1 - x_{i,t}) + SOC_{i,t}^{\mathrm{init}} \cdot x_{i,t} \; \forall i \in I, t \in T \tag{2}$$

$$soc_{i,t-1} + soc_{i,t}^{\mathrm{short}} \geq BC_g^{\max} \cdot S_{i,g} \cdot x_{i,t} \; \forall i \in I, g \in G, t \in T \tag{3}$$

$$\sum_{(i\in I)} S_{i,g} \cdot x_{i,t} + bat_{g,t}^{\mathrm{short}} = N_{g,t} \;\; \forall g \in G, t \in T \tag{4}$$

$$em_t^{\mathrm{buy}} - em_t^{\mathrm{sell}} = \sum_{(i\in I)} \left(bat_{i,t}^{\mathrm{chg}} - bat_{i,t}^{\mathrm{dsg}} \right) \;\; \forall t \in T \tag{5}$$

$$0 \leq bat_{i,t}^{\mathrm{chg}} \leq (1 - x_{i,t}) P_i^{\max} \;\; \forall i \in I, t \in T \tag{6}$$

$$0 \leq bat_{i,t}^{\mathrm{dsg}} \leq (1 - x_{i,t}) P_i^{\max} \;\; \forall i \in I, t \in T \tag{7}$$

$$\sum_{(g\in G)} BC_g^{\min} \cdot S_{i,g} \leq soc_{i,t} \leq \sum_{(g\in G)} BC_g^{\max} \cdot S_{i,g} \quad \forall i \in I, t \in T \tag{8}$$

$$\sum_{(g\in G)} BC_g^{\min} \cdot S_{i,g} \leq soc_{i,t}^{\mathrm{short}} \leq \sum_{(g\in G)} BC_g^{\max} \cdot S_{i,g} \quad \forall i \in I, t \in T \tag{9}$$

$$bat_{i,t}^{\mathrm{dsg}} \leq P_i^{\max} \cdot a_{i,t} \;\; \forall i \in I, t \in T \tag{10}$$

$$bat_{i,t}^{\mathrm{chg}} \leq P_i^{\max} (1 - a_{i,t}) \;\; \forall i \in I, t \in T \tag{11}$$

$$em_t^{\mathrm{sell}} \leq M \cdot c_t \;\; \forall t \in T \tag{12}$$

$$em_t^{\mathrm{buy}} \leq M (1 - c_t) \;\; \forall t \in T \tag{13}$$

$$soc_{i,t=|T|} = SOC_{i,t=0}^{\mathrm{init}} \;\; \forall i \in I \tag{14}$$

$$\frac{soc_{i,t}^{\mathrm{short}}}{\sum_{(g\in G)} S_{i,g} \cdot BC_g} = \sum_{(d\in D)} L_{i,t,d} \;\; \forall i \in I, t \in T \tag{15}$$

$$\beta_{i,t} = \sum_{(d\in D)} k_d \cdot L_{i,t,d} \;\; \forall i \in I, t \in T \tag{16}$$

$$0 \leq L_{i,t,d} \leq L_d^{\max} \;\; \forall i \in I, t \in T, d \in D. \tag{17}$$

In (1), the first term is the revenue collected from customers for swapping priced at BSR. The binary variable $x_{i,t}$ is equal to 1 if battery i is swapped at the beginning of time period t, and 0 otherwise. The second term is the energy purchased and sold in the DA market at price λ_t^{DA}. The objective function also considers the BSS's inability to supply battery demand $bat_{g,t}^{\mathrm{short}}$, which is penalized at the cost of value-of-customer-dissatisfaction, $VoCD$. The last term is the discount $\beta_{i,t}$ given on the revenue BSR collected from consumers for a battery swap if a 100% charged battery cannot be supplied.

The $VoCD$ is a monetary value the BSS business places on the inability to supply a customer with a battery. This value is determined by performing research studies and surveys of its customers. A parallel can be drawn between the $VoCD$ and the value-of-lost-load ($VoLL$), which is used in unit commitment models, e.g., [22]. $VoLL$ represents the value customers place on the loss of 1 MWh of supply, which is dependent on interruption duration, time of day, location, and cause of interruption [23]. The BSS has two criteria when estimating the $VoCD$: i) the monetary loss of swapping revenue that could have been obtained at present and potentially in the future from the dissatisfied customer, and ii) the monetary loss due to new customer acquisition. However, the estimation of $VoCD$ is beyond the scope of this work.

Equation (2) is the energy state-of-charge $soc_{i,t}$ of each battery i. This energy state-of-charge takes into account the state-of-charge at the previous time period, the charging power $bat_{i,t}^{\mathrm{chg}}$, discharging power $bat_{i,t}^{\mathrm{dsg}}$, and efficiencies. Thus, in (2), if $x_{i,t}$ is equal to 1, the $soc_{i,t}$ is not updated. Instead, the battery is swapped with a different one with the incoming energy state-of-charge, $SOC_{i,t}^{\mathrm{init}}$. Constraint (2) performs multiplication of continuous and binary variables, which are linearized with the approach discussed in [24]. Constraint (3) captures the battery swapping mechanics. If battery i is swapped to an EV, the state-of-charge should be at the maximum battery capacity $BC_g^{\max}$, for the battery group g it belongs to. Binary parameter $S_{i,g}$ is used to identify the appropriate battery group g, which battery i belongs to. If the maximum battery capacity cannot be met, the remainder is captured in the energy shortage variable $soc_{i,t}^{\mathrm{short}}$. In (4), The BSS seeks to meet the battery demand $N_{g,t}$. Battery shortage is captured in variable $bat_{g,t}^{\mathrm{short}}$, which is penalized in (1). To meet the demand, energy needs to be purchased from the market and the excess can be sold back if this results in profit for the BSS, as shown in (5).

Each battery has a maximum charging/discharging power limit $P_i^{\max}$, enforced by constraints (6) and (7). The batteries energy state-of-charge and energy state-of-charge shortage are constrained by maximum and minimum battery capacities in constraints (8) and (9). The energy state-of-charge ceiling is below the nominal value in order to avoid the risk of setting the battery on fire, and the minimum avoids rapid degradation of the battery, as explained in [25]. Constraints (10) and (11) disallow charging and discharging to occur simultaneously at any given period of time, while (12) and (13) disallow activating the purchasing and selling variables simultaneously. Constraint (14) ensures the total stored energy is the same as it was at the beginning of the day.

If the BSS cannot manage to fully charge a battery that potentially could be swapped to a customer, it has two alternatives. The first is to offer a discount on the price of the battery swap, BSR. The second alternative is not to swap the battery incurring the $VoCD$ penalty. The discounts are modeled as a piecewise linear curve in Fig. 2 and they increase as the battery

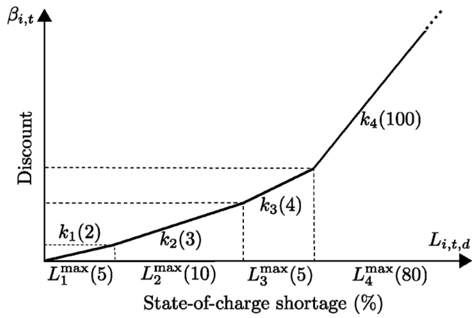

Fig. 2. Piecewise discount function. The values in $(\cdot)$ are used in the case studies in Section IV.

energy shortage increases. The piecewise curve needs to incorporate the discount due to the energy shortage in the swapped battery and the perceived inconvenience of the customer having to return to the BSS sooner due to the energy shortage. This is modeled through constraints (15)–(17). Constraint (15) calculates the normalized energy state-of-charge shortage and distributes it among the discrete energy state-of-charge shortage segments $L_{i,t,d}$. The discount $\beta_{i,t}$ in (16) is calculated as the product of the slope of each discount segment k_d and the respective shortage segment L_d. Lastly, constraint (17) limits the state-of-charge shortage segments to their preset limits $L_d^{\max}$. An example is shown in Fig. 2. For the first 5% ($L_1^{\max}$) of the missing energy, the BSS gives up to 10% discount ($k_1 \cdot L_1^{\max}$) on the price of the battery swap, BSR. For the next 10% ($L_2^{\max}$) of the missing energy, the BSS gives up to an additional 30% discount ($k_2 \cdot L_2^{\max}$). This rationale follows for the remaining segments.

C. Demand Uncertainty With Inventory Robust Optimization

At the DA stage, battery demand intervals $[N_{g,t}^{\min}, N_{g,t}^{\max}]$ represent the bounds of the expected battery demand for the next day. If the battery demand is high, the BSS will need to resort to purchasing energy in the volatile real-time market to meet the demand. Furthermore, the BSS might need to offer discounts for non-fully charged batteries or it might even be unable to supply some customers. On the other hand, if the battery demand realization is low, the excess electricity can either be sold in the real-time market or stored in the batteries for later use. However, having a battery stock fully charged would prevent the BSS to take advantage of possible low prices in the real-time market. This uncertainty is managed using the robust inventory theory [17]. In such approach, a sub-optimal solution is obtained if the demand realizations are within the defined interval that belongs to $[N_{g,t}^{\min}, N_{g,t}^{\max}]$, where $N_{g,t}^{\max} = N_{g,t}^{\min} + \Delta N_{g,t}$ and $\Delta N_{g,t}$ is the battery demand deviation. The DA model (1)–(17) extended to account for demand uncertainty is formulated as follows:

$$\text{maximize} \quad BSR \sum_{(t\in T)} \sum_{(i\in I)} x_{i,t} - \sum_{(t\in T)} \lambda_t^{\mathrm{DA}} \left(em_t^{\mathrm{buy}} - em_t^{\mathrm{sell}} \right)$$

$$- VoCD \sum_{(t\in T)} \sum_{(g\in G)} bat_{g,t}^{\mathrm{short}} - BSR \sum_{(t\in T)} \sum_{(i\in I)} \beta_{i,t} - \sum_{(g\in G)} y_g \quad (18)$$

subject to

$$\text{Constraints } (2),\ (3),\ (5)\text{-}(17) \quad (19)$$

$$\sum_{(i\in I)} S_{i,g} \cdot x_{i,t} + bat_{g,t}^{\mathrm{short}} = N_{g,t}^{\min} + bat_{g,t}^{\mathrm{add}} \ \ \forall g \in G, t \in T \quad (20)$$

$$y_g \geq VoCD \left(- \sum_{(t\in T)} bat_{g,t}^{\mathrm{add}} + \Gamma_g \cdot q_g + \sum_{(t\in T)} r_{g,t} \right) \quad \forall g \in G \quad (21)$$

$$q_g + r_{g,t} \geq \Delta N_{g,t} \ \ \forall g \in G, t \in J_g. \quad (22)$$

The robust objective function is shown in (18). The last term represents the penalty cost y_g imposed by the inventory RO for each battery group g. Equation (20) ensures the minimum battery demand ($N_{g,t}^{\min}$) is met. Variable $bat_{g,t}^{\mathrm{add}}$ represents the additional number of batteries charged on top of $N_{g,t}^{\min}$. Constraint (21) determines the worst time periods in which additional battery demand could materialize. Parameter Γ_g is used to control the level of robustness of the solution taking values in $[0, |J_g|]$, where $J_g = \{t | \Delta N_{g,t} > 0\}$. If $\Gamma_g = 0$, no deviations are considered and the solution is equivalent to the deterministic one, whereas if $\Gamma_g = |J_g|$, deviations at all time periods are considered, acquiring the most conservative solution (i.e., demand at all time periods is equal to $N_{g,t}^{\max}$). Variables q_g and $r_{g,t}$ are non-negative auxiliary variables used to account for the known demand bounds. Variable $r_{g,t}$ is incorporated in constraints (21)–(22) as the dual variable of constraint (4) and q_g is used to preserve linearity. A detailed explanation of how to obtain the inventory RO problem is given in [17].

D. Price Uncertainty With Multi-Band Robust Optimization

The BSS can interact with electricity markets by purchasing energy when it is inexpensive and injecting it back when the price is high, at a profit. Profit opportunities may be missed or the BSS could incur losses if uncertainty of market prices is not appropriately accounted for. To hedge against this uncertainty, the multi-band RO approach is used [16].

In [16], deviations in prices are modeled within the range $[\lambda_t^{\min}, \lambda_{t,b}^{\max}]$, where $\lambda_{t,b}^{\max} = \lambda_t^{\min} + \Delta\lambda_{t,b}$ and $\Delta\lambda_{t,b}$ is the market price deviation for each band b. Unlike inventory RO used for the battery demand uncertainty, the multi-band approach uses multiple deviation bands controlled by robustness parameter θ_b, which takes values in $[0, |U_b|]$, where $U_b = \{t | \Delta\lambda_{t,b} > 0\}$. However, the total number of deviations over the optimization horizon must be $\sum_{(b\in B)} |\theta_b| \leq |T|$. As an example, Fig. 3 shows three uncertainty bands. The band $\lambda_{t,5\%}^{\max}$ represents all the uncertainty materializations that are 5% above $\lambda_t^{\min}$. Similar rationale applies to $\lambda_{t,10\%}^{\max}$ and $\lambda_{t,15\%}^{\max}$. The formulation of the model that manages price uncertainty using multi-band RO is

$$\text{maximize } BSR \sum_{(t\in T)} \sum_{(i\in I)} x_{i,t} - \sum_{(t\in T)} \lambda_t^{\min} \left(em_t^{\mathrm{buy}} - em_t^{\mathrm{sell}} \right)$$

$$- VoCD \sum_{(t\in T)} \sum_{(g\in G)} bat_{g,t}^{\mathrm{short}} - BSR \sum_{(t\in T)} \sum_{(i\in I)} \beta_{i,t}$$

$$- \sum_{(b\in B)} \theta_b \cdot v_b - \sum_{(t\in T)} z_t \quad (23)$$

subject to

$$\text{Constraints } (2)\text{-}(17) \quad (24)$$

$$v_b + z_t \geq \Delta\lambda_{t,b} \cdot em_t^{\mathrm{buy}} \ \ \forall b \in B, t \in U_b. \quad (25)$$

The objective function (23) includes additional variables v_b and z_t, and parameter θ_b, which controls the robustness of the solution. If $\theta_b = 0$, the effect of price deviations is ignored and the solution is deterministic. On the other hand, for $\theta_b = |U_b|$

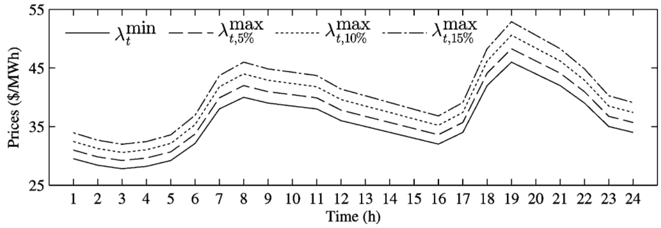

Fig. 3. DA price deviations with multiple bands.

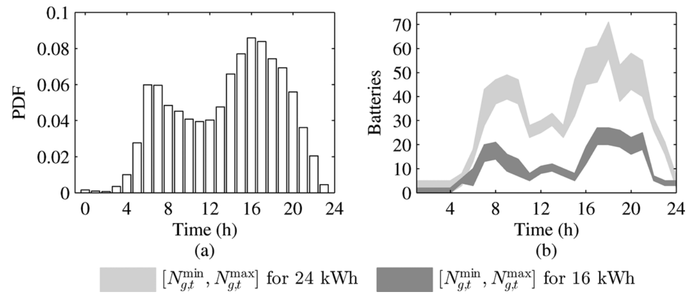

Fig. 4. Distribution of demand at BSS in (a), and uncertainty bounds in (b).

all price deviations are considered, which results in the most conservative solution. Equation (25) ensures feasibility for any deviation $\Delta\lambda_{t,b}$. The non-negative variable z_t is the dual variable of the DA objective function (1) and the non-negative variable v_b is used for linear equivalency. A detailed explanation of how to obtain the multi-band RO problem is discussed in [16].

E. Battery Degradation Costs

Charging cycles reduce batteries' lifetime. Since the BSS incurs all costs related to batteries, it needs to incorporate the costs resulting from battery degradation. In this work, degradation characteristic is highly sensitive to the number of cycles and virtually insensitive to the depth-of-discharge, as explained in [26]. However, other chemistries with higher sensitivity to the depth-of-discharge can be modeled as proposed in [26]. The formulation of the BSS model that manages battery degradation is

$$\begin{aligned}\text{maximize} \quad & BSR\sum_{(t\in T)}\sum_{(i\in I)} x_{i,t} - \sum_{(t\in T)} \lambda_t^{\mathrm{DA}}\left(em_t^{\mathrm{buy}} - em_t^{\mathrm{sell}}\right)\\ & - VoCD\sum_{(t\in T)}\sum_{(g\in G)} bat_{g,t}^{\mathrm{short}}\\ & - BSR\sum_{(t\in T)}\sum_{(i\in I)}\beta_{i,t} - \sum_{(t\in T)}\sum_{(i\in I)} C_{i,t}^{\deg} \end{aligned}\tag{26}$$

subject to

$$\text{Constraints (2)-(17)}\tag{27}$$

$$C_{i,t}^{\deg} = \left|\frac{m_i}{100}\right| \frac{bat_{i,t}^{\mathrm{chg}} + bat_{i,t}^{\mathrm{dsg}}}{\sum\limits_{g\in G} S_{i,g}\cdot BC_g} C_i^{\mathrm{bat}} \quad \forall i\in I, t\in T.\tag{28}$$

The objective function (26) explicitly includes the cost of degrading the battery $C_i^{\deg}$, due to charging/discharging cycles undergone. $C_i^{\deg}$ is calculated in (28), where m_i is the slope of the linear approximation of battery life as a function of the number of cycles, and C_i^{bat} is the cost of the battery. The value of m_i is approximated from manufacturer datasheets [27].

F. Complete Day-Ahead Model

The complete DA model that incorporates battery demand uncertainty, market price uncertainty, and battery degradation costs is as follows:

$$\begin{aligned}\text{maximize}\ & BSR\sum_{(t\in T)}\sum_{(i\in I)} x_{i,t} - \sum_{(t\in T)} \lambda_t^{\min}\left(em_t^{\mathrm{buy}} - em_t^{\mathrm{sell}}\right)\\ & - VoCD\sum_{(t\in T)}\sum_{(g\in G)} bat_{g,t}^{\mathrm{short}} - BSR\sum_{(t\in T)}\sum_{(i\in I)}\beta_{i,t}\\ & - \sum_{(g\in G)} y_g - \sum_{(b\in B)}\theta_b v_b - \sum_{(t\in T)} z_t - \sum_{(t\in T)}\sum_{(i\in I)} C_{i,t}^{\deg} \end{aligned}\tag{29}$$

subject to

$$\text{Constraints (2), (3), (5)-(17), (20)-(22), (25), (28).}\tag{30}$$

IV. Case Study

The proposed approach is applied to two case studies over a 24 hour period. The charging/discharging efficiencies are 90% with maximum charging/discharging power of 3.3 kW. The energy state-of-charge variable ($soc_{i,t}$) is bounded within 15% and 95% of the nominal capacities in order to protect the batteries [25]. The initial energy state-of-charge ($SOC_{i,t=0}^{\mathrm{init}}$) of the batteries are uniformly randomized between 15% and 95%, and the incoming energy state-of-charge ($SOC_{i,t>0}^{\mathrm{init}}$) is randomized within 30% and 60% of the nominal capacity. Parameter $SOC_{i,t=0}^{\mathrm{init}}$ represents the energy stored in each battery in the BSS stock at the beginning of the day, whereas $SOC_{i,t>0}^{\mathrm{init}}$ represents the energy in the customer's battery when they arrive to the BSS for swapping. The BSR is set to \$70, as used in [13], and the $VoCD$ is \$200. The piecewise discount curve is designed with four segments: 1) $k_1 = 2$ and $L_1^{\max} = 5\%$, 2) $k_2 = 3$ and $L_2^{\max} = 10\%$, 3) $k_3 = 4$ and $L_3^{\max} = 5\%$, and 4) $k_4 = 100$ and $L_4^{\max} = 80\%$, and is shown in Fig. 2. In order to represent a typical weekday, the historical data of every Thursday in the period January-March 2013 from the PJM (Pennsylvania-New Jersey-Maryland) market [28] was used. The typical weekday curve was obtained using the K-means clustering approach [29] to identify a price profile that best characterizes the data. The upper bound prices $\lambda_{t,b}^{\max}$ are proportional to the derived typical weekday curve which is equivalent to $\lambda_t^{\min}$, as shown in Fig. 3. The arrival times for swapping are assumed to follow the probability distribution function (PDF) shown in Fig. 4(a), which is derived from [30]. This assumption is justifiable since there is no historical data available for any existing BSS. The demand range $[N_{g,t}^{\min}, N_{g,t}^{\max}]$ for battery group 24 kWh is shown in Fig. 4(b) in light grey, and for 16-kWh batteries in dark grey. The model is a mixed-integer linear programming (MILP) model implemented in GAMS 24.0 and solved using CPLEX [31].

A. Small Battery Stock With a Single Battery Type

This study assesses the individual effects of price uncertainty, demand uncertainty, and battery degradation. A single battery group is used with a stock of 200 batteries with nominal capacities of 24 kWh, as shown in Fig. 4(b) in light grey area.

1) Effect of Battery Degradation: By varying the battery capacity cost from 500 (cost in 2012) to 100 \$/kWh, accounting for the advancement of battery technologies as time progresses, the effect on B2G and B2B is studied. The degradation model

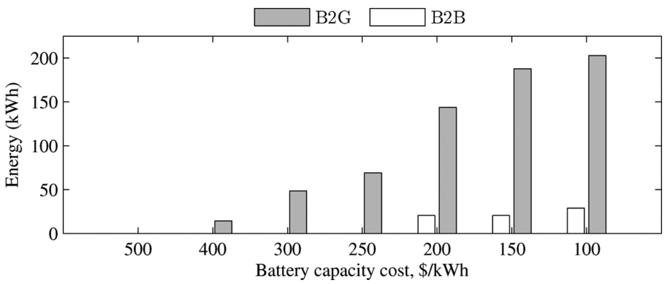

Fig. 5. Impact on B2G and B2B as battery capacity cost decreases.

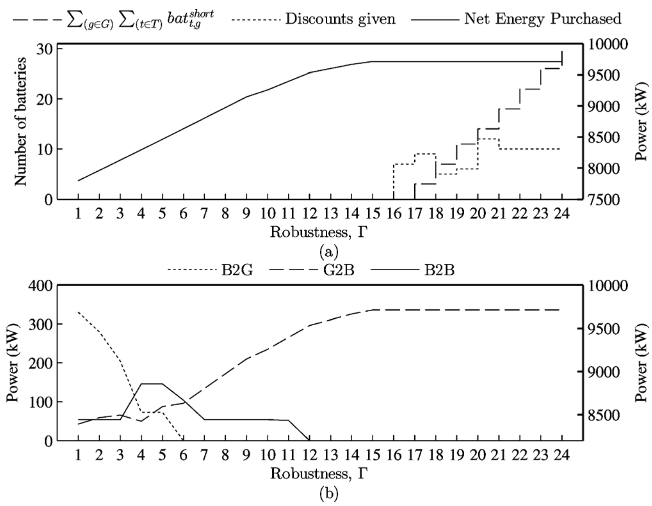

Fig. 6. (a) Effect on battery shortage, discounts, and net energy purchased, and (b) effect on B2G, G2B, and B2B services. B2G and B2B are referred to the left-side-axis and G2B to the right-side-axis.

of the batteries is linear and represented by (28) with slopes $m_i = [-0.0158, -0.0009]$. The lower slope is the linear approximation of the 2012 technology [24], and the higher slope represents a technological improvement in the battery cycle-life of three times the current value. Fig. 5 shows the energy transfer in B2G and B2B modes for different values of battery capacity cost. When the battery capacity cost is 500 $/kWh, the battery performs no B2G or B2B. However, these services become attractive as the battery capacity cost decreases, reaching a maximum for the lowest battery capacity cost. If B2G and B2B are not present, the BSS purchases only the electricity needed to supply the battery demand. B2B is significantly lower than B2G since B2B requires a battery to discharge in order to charge another battery. This causes degradation of both batteries and thus, the overall cost increases. However, B2G supplies electricity to the system by discharging a battery and thus, only degrading a single battery during the optimization horizon. Fig. 5 shows that as the battery technology improves, the provision of services to the grid, and internal energy transfers in B2B result in economic benefits to the BSS.

2) Effect of Uncertainty in Battery Demand: As discussed in Section III-C, the robustness parameter Γ controls the level of robustness in the objective. Fig. 6(a) shows the battery shortage, discounts given (dashed line), and net energy purchased (solid line) to charge batteries as a function of Γ. Fig. 6(b) shows the total energy in B2G, G2B, and B2B modes as a function of Γ. The net energy purchased is defined as the energy required to meet the battery demand over the optimization horizon.

In Fig. 6(a), it can be seen that as Γ increases, the net energy purchased to charge batteries increases, because the BSS needs to schedule additional batteries for the worst-case realizations (within the ranges shown in Fig. 4(b)) without having to resort to offer discounts, or incur $VoCD$. This figure shows that the net energy purchased saturates after $\Gamma = 15$, since the BSS has insufficient stock to offer fully charged batteries, and thus needs to give discounts and even is unable to serve all the customers. The battery stock is the limiting factor because additional batteries cannot be scheduled above $\Gamma = 15$.

Fig. 6(b) shows B2G, G2B and B2B as a function of Γ. This figure shows that B2G decreases until $\Gamma = 6$, because additional batteries are scheduled in order to cope with the uncertainty. This results in energy that used to be sold to the grid to be used for swapping purposes. On the other hand, B2B reaches the maximum for $\Gamma = 4$ and $\Gamma = 5$ because the demand increase requires more energy. However, before resorting to buying energy from the grid, B2B is a less expensive option. This is because stand-by batteries pre-charge during the low-price periods, and supply other batteries in the BSS stock when they require energy during the high-price periods. For Γ values from 6 to 12, B2B reduces because the batteries that used to pre-charge are now required to be swapped. Thus, they store their pre-charged energy rather than transferring to other batteries. For $\Gamma \geq 12$, B2G and B2B do not occur because the additional batteries scheduled for swapping has increased up to the point where the BSS cannot exploit the advantages of selling and transferring energy, but has to store energy for swapping.

Battery demand uncertainty reduces the attractiveness of B2G and B2B services at the benefit of protection against the worst-case demand realizations. The BSS needs to decide what level of robustness is desired in the DA scheduling by weighing the benefits of B2G and B2B against incurring discounts and $VoCD$ if it is not able to meet the demand.

3) Effect of Uncertainty in Market Prices: In general, the BSS tries to purchase electricity during the lowest-price periods. However, with market price uncertainty, electricity purchases span over multiple time periods to protect against price deviations. The robustness parameters $\theta_{10\%}$ and $\theta_{15\%}$, are varied between $[0, |U_b|]$ for price deviations of 10% and 15% above $\lambda_t^{\min}$ as shown in Fig. 3. The variations, however, need to follow the rule $|\theta_{10\%}| + |\theta_{15\%}| \leq |T|$ [16]. The model is solved for these combinations of robustness parameters and the effect on B2G and B2B are shown in Fig. 7. Quantities in Fig. 7(a) are normalized to the maximum energy in B2G mode of 463.16 kWh, and in Fig. 7(b) to the maximum energy in B2B mode of 120.60 kWh.

As an example, if the BSS decides to protect against eight deviations in the 10% band and four deviations in 15% band, i.e., $\theta_{10\%} = 8$ and $\theta_{15\%} = 4$, the B2G would be approximately 0.81–0.90 p.u and B2B would be 0.56–0.65 p.u. In Fig. 7(a), B2G decreases as the robustness parameter is increased. However, B2G decreases faster to its minimum of 0.45–0.35 p.u. for increases in $\theta_{15\%}$ as compared to $\theta_{10\%}$. This is the case because 15% price deviations are associated to larger price increments as compared to 10%. There are two reasons for the decreasing trend of B2G as a result of the price uncertainty. First, B2G requires excess electricity to be purchased and stored in batteries. By decreasing B2G, the amount of electricity purchased decreases, minimizing the potential impact of high-price materializations. Second, the large electricity purchases that were concentrated in the low-price periods, in

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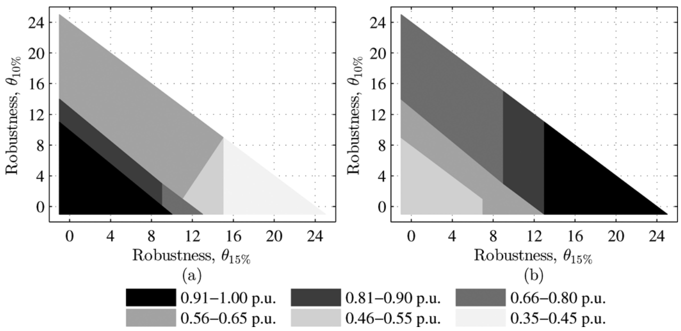

Fig. 7. Effect of price uncertainty on the energy injected in B2G (a) and B2B (b) in p.u. (i.e., normalized over kWh).

order to minimize costs, are spread over multiple periods. This causes higher-price periods to purchase rather than sell energy to cope with the uncertainty.

Since the RO approach spans the energy purchases over time, resulting even in purchases during the high-price periods, B2B is an economic alternative to the expensive market purchases. By exercising B2B, pre-charged batteries supply the batteries that require energy for swapping. From Fig. 7(b), it can be seen that if the robustness parameters are set to zero ($\theta_{10\%} = \theta_{15\%} = 0$), B2B is at its minimum of 0.46–0.55 p.u. As the robustness parameters increase, B2B increases as well. As in the B2G case, a higher increase in B2B occurs for 15% price deviations as compared to the 10% price deviations.

B. Large Battery Stock With Two Battery Types

In the large-scale case study, 16-kWh and 24-kWh battery groups are used with 100 and 200 battery stock, respectively, as shown in Fig. 4(b). From Fig. 5, battery capacity cost is set to 200 $/kWh to denote a trend towards cheaper batteries in the future, where the benefits of B2G and B2B will be more evident. This study uses the complete DA model, as described in Section III-F.

To determine the optimal combination of RO parameters, Monte Carlo (MC) simulations were performed [32]. The DA schedules for all combinations of Γ_g for 16- and 24-kWh batteries, and θ_b for 10% and 15% price deviation bands are simulated against MC trials. The number of MC trials is set to the min $\{1000, N^{\mathrm{MC}}\}$, where N^{MC} is the number of MC trials required to ensure a 95% confidence of an error less than 1% [32]. All combinations of demand robust parameters Γ_g and price robust parameters θ_b were simulated against 32 market price and 32 battery demand profiles, totaling 1024 MC trials.

In order to create price profiles, the difference between the typical curve, shown in Fig. 3, and the historical data from the PJM market for each time period was calculated, which represents the error. Next, the cumulative distribution function (CDF) of the error is obtained for each time period considering the inter-hour correlation. Finally, price profiles are created using the error distribution. The distribution function in Fig. 4(a) is used to create the demand profiles.

Fig. 8 shows the normalized cumulative distribution functions (CDFs) of the BSS profits for different combinations of robustness parameters. Although the MC simulations were performed for all combination of parameters, only selected CDFs are shown for clarity. Each MC trials' DA profit is obtained for the robustness combination without uncertainty, i.e., $\Gamma_g = 0$ and $\theta_b = 0$, and these profits are used as the base-case, in which all other combination MC trials' profits are normalized against.

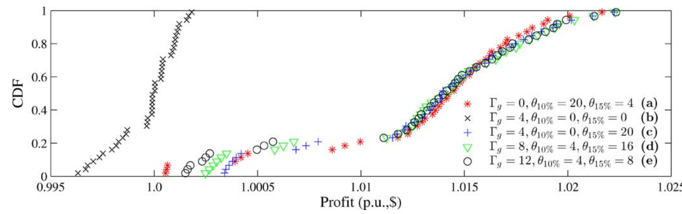

Fig. 8. CDF for combination of robustness parameters for price and demand.

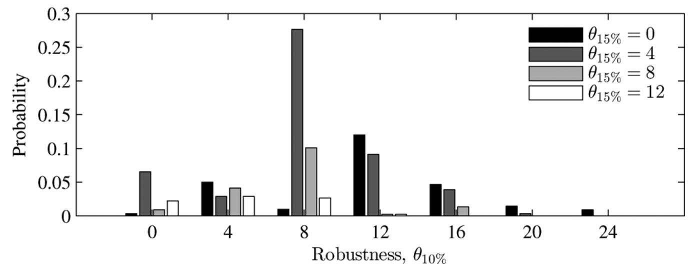

Fig. 9. Probability of resulting in the highest overall profit for all combinations of $\theta_{10\%}$ and $\theta_{15\%}$.

In general, the combination that yields the right-most CDF performs best, since it obtains the highest profits. From Fig. 8, the lower tails of the right-most group of CDFs show differences in profit, where case (c) (labelled in Fig. 8) yields the highest. However, there is no distinct CDF that can be determined as the optimal combination. This is the case because the right-most CDFs do not yield a large difference in profit for the majority of the MC trials, apart from the lower-tail. Since the right-most CDFs are similar, the BSS has less risk in determining an optimal combination of robust parameters for battery demand uncertainty. Therefore, to analyze the remainder of the results, Γ_g is chosen as 4 for both the 16- and 24-kWh battery groups.

On the other hand, if price uncertainty is ignored as in the case labeled (b) in Fig. 8, the CDF is the one that has lowest profits (left-most) with some trials' profit below 1 p.u. Therefore, price uncertainty parameters (θ_b) need to be properly chosen otherwise a decrease in profits may occur. To determine θ_b combinations, MC simulations are performed but this time each MC trial is characterized by a fixed demand profile ($N_{g,t}^{\min}$ from Fig. 4(b)), with different price profiles. The combination of robust parameters that result in the highest overall profit was identified for each MC trial. Fig. 9 shows the probability of yielding the highest overall profit for different combinations of robustness parameters $\theta_{10\%}$ and $\theta_{15\%}$. The chances of incurring the highest profit for $\theta_{10\%} = 0$ and $\theta_{15\%} = 0$ is less than 1%, because no protection against price uncertainty is used. The optimal combination of $\theta_{10\%}$ and $\theta_{15\%}$ is 8 and 4, which results in probability of 27% for obtaining the highest profits. This combination of price robustness parameters is used while analyzing the remainder of the results.

At the DA stage, the BSS determines its energy bidding and offering schedules which are equivalent to G2B and B2G mode, as shown in Fig. 10(a) for the deterministic case when battery demand and market price uncertainty are not included. If the market price is high, the BSS sells electricity whereas during the low-price periods it purchases electricity. Fig. 10(b) shows the energy transfer between batteries in B2B mode, which occurs during the high-price periods. The figure shows that B2B

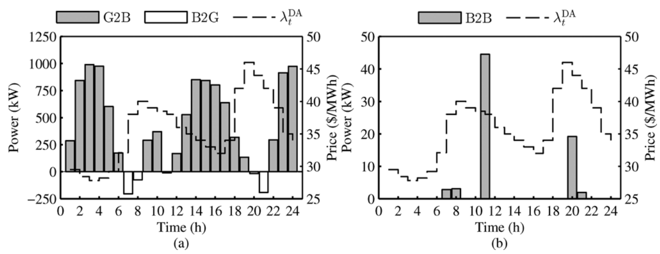

Fig. 10. G2B and B2G (a), and B2B (b) services in the deterministic case.

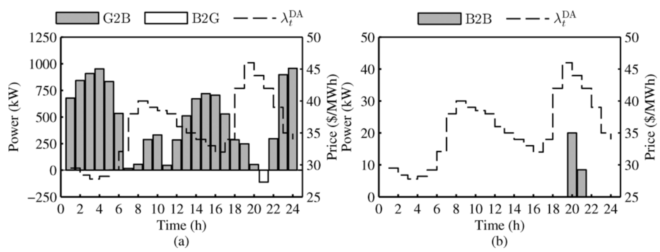

Fig. 11. G2B and B2G (a), and B2B (b) services in the uncertainty case.

can occur simultaneously with B2G. Since battery demand and market prices are high during periods in which B2B occurs, the BSS uses stand-by batteries to discharge and charge other batteries. This way, they can be swapped and the remaining energy is sold in the market in B2G mode. B2B mode counteracts electricity to be purchased in G2B mode during the high-price periods. The total electricity scheduled in B2G mode is 499.40 kWh, and in B2B mode is 71.51 kWh.

With market price and battery demand uncertainty included, as shown in Fig. 11, the BSS is more conservative by purchasing more electricity to supply the uncertain battery demand in G2B mode. Battery demand uncertainty decreases B2G services because the energy that was used for pre-charging batteries to be sold in later periods is instead being used to supply the battery demand. Market price uncertainty also decreases B2G because it limits the electricity that can be purchased at a competitive price during low-price periods. Both of these effects can be seen in Fig. 11(a) where the total energy purchased increases caused by the uncertain battery demand. However, the purchasing is spread between periods in order to minimize losses caused by realization of high prices.

Battery degradation further limits B2G since this service comes at a cost of additional charging in order to discharge later on, which degrades the battery life. From Fig. 11(a), B2G decreases by 77% to 112.56 kWh as compared to the deterministic case. Due to market price uncertainty, B2B decreases by 60% to 28.46 kWh in Fig. 11(b) as compared to the deterministic case. The uncertainties decrease B2G and B2B, however it also protects against the worst-case materializations of prices and demand.

In the deterministic case, the total profit over the 24-h period is \$55 648 and costs are \$352.20, of which \$323.44 correspond to electricity costs and \$28.76 to degradation costs. When considering uncertainty, the profit is \$55 609, while costs are \$421.63, of which electricity costs are \$361.1 and degradation costs are \$30.26. The remainder totaling to \$30.27 indicates the cost of including uncertainty in the model. The profit is lower and losses are higher when considering the battery demand and price uncertainty because the BSS purchases more energy and minimizes the purchases during the low-price periods.

The losses the BSS incurs do not include any discounts in the deterministic and uncertain cases, because the battery demand is met and the excess energy is sold in the market in B2G mode. If cases occur where the battery demand increases, the amount of energy in B2G mode would decrease until only G2B mode takes place at all time periods. Any further increase in the battery demand will require discounts to be given to customers, since the BSS is purchasing at maximum capacity of all the batteries in G2B mode. In the worst case, when even discounts cannot satisfy the demand, then it is because battery stock is too small and $VoCD$ would be incurred by the BSS for each battery not supplied.

V. Conclusion

A detailed operating model of the Battery Swapping Station (BSS) is presented in this paper. The BSS can schedule batteries to operate in G2B, B2G, or B2B modes. The model employs robust optimization techniques to manage uncertainty of electricity prices and battery demand. In addition, the effect of battery degradation due to the charging/discharging cycling is taken into account as well as their economic losses.

The proposed model and the results that can be obtained by it will: 1) help inform stakeholders on the design and operation of BSS stations, 2) allow more efficient short-run and long-run market decisions that can exploit storage capabilities of the BSS, and 3) enhance the environmental sustainability of the power sector by allowing further introduction of renewable energy sources.

Results show that accounting for battery demand uncertainty, electricity price uncertainty, and battery degradation costs, decreases B2G and B2B services. Electricity price uncertainty has a major impact on the profits of the BSS and it needs to be managed properly in order to avoid poor economic performance. On the other hand, the impact of battery demand uncertainty is less severe, but proper management of this uncertainty results into better operating strategies for the BSS. The order of priority for the services that the BSS can perform are dependent on the excess energy, in addition to battery swapping needs, obtained in G2B mode. Then, depending on the market conditions (e.g., high energy prices), the BSS schedules some of the batteries to transfer energy to other batteries in B2B mode. Lastly, if there is an economic benefit, the BSS sells the excess energy in B2G mode to the market.

As for the battery degradation, current battery costs (data in 2012) are still too high to allow BSS to exploit B2B and B2G services. This is expected to change when battery technology grows more mature, and prices and degradation resilience improves. It is important for the BSS to consider these features in its business and operating models in order to maximize economic performance.

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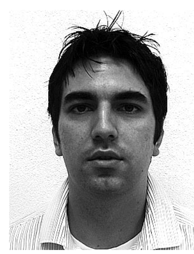

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